

INFO 2950: Phase I

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Project workspace (GitHub): <https://github.com/Harashima-CS/info2950-project>.

Theme

All three of our ideas use large, accessible chess datasets. We chose chess because every game played on sites like chess.com and lichess is automatically recorded. These records include extensive data, including each player's remaining time at every move and their estimated skill (Elo score). We can therefore ask how skill and time affect decision quality.

IDEA 1: How does the probability of blundering depend on how much time a player has? How does that relationship change between bullet, blitz, rapid, and classical time modes?

Source: Lichess public API (<https://lichess.org/api>). This dataset is free, with no API key needed, and includes rated Lichess games, including timestamps, move time, time mode (15m game vs 30 vs 1hr...), players' ratings, and engine evaluation. We plan to compare the relationship between the clock and the number of blunders across several time controls. We can export tournament and user games, including the engine eval and timestamps. Lichess' recurring arena tournaments are a good way to observe multiple games of one time mode across a wide rating range. We can also use the Lichess open database (<https://database.lichess.org>), but we would probably use it more to check the API dataset than as the primary data source. One problem we noticed is that only ~6% of games have engine evals. I think, however, that we could run the other games through Stockfish ourselves via a Python library. The main advantage of this

dataset is that it's stored automatically, so we can assume it's mostly trustworthy because it's not self-reported. This applies to ideas 2 and 3 as well. Our main concern, besides how we can get engine evals for every game, is interference from cheating/engine use during games that could skew our data.

IDEA 2: At what rating do opening traps and unsound “trick” openings stop working? OR within one opening, like The Italian Game, how do players’ move choices change based on rating?

Source: Lichess Opening Explorer API (<https://lichess.org/api#tag/Opening-Explorer>). This dataset includes positions from many opening lines, including how often each move is played and whether they commonly lead to a win/draw/loss, as well as the elo (rating). By using the engine to evaluate each position, we could compare how objectively “good” an opening is vs. how it actually performs. We can investigate how well traps like the Fried Liver or England gambit work, and at what rating they are no longer effective. Using the API requires a personal access token, but its free for any Lichess account to create. We can combine this with engine evaluations via Lichess’ cloud-eval API or the open evaluations database (<https://database.lichess.org/#evals>). The main advantage of this dataset is size, because even niche openings have thousands of games available at lower ratings. One disadvantage is that there might not be enough data available for niche openings in higher elo games.

IDEA 3: For top online players, does losing a game increase the likelihood of losing the next?

Source: Chess.com Published-Data API (<https://www.chess.com/news/view/published-data-api>).

This dataset contains every public Chess.com game (by player). It includes date, time, ratings, result, opening, Chess.com's "accuracy" score (how close the players' moves were to optimal), as well as a full list of moves. We can focus specifically on one (or a few) players like Hikaru Nakamura and use the timestamps to determine how his playing is affected by previous results. The chess.com API is free and returns a player's games by month. Even just Hikaru's games on Chess.com gives us a large dataset. We noticed is that Chess.com's accuracy score is sometimes missing, so we would need to mainly rely on game results. The main advantage of this dataset is that it comes directly from Chess.com's servers. One concern is that studying only one player means our results describe Hikaru rather than players in general, and his casual or handicapped games would need to be filtered out.

Questions:

- For idea 1, does using the Lichess database for most games but using the Stockfish API to evaluate them ourselves count as more than one dataset? Or is it better to collect everything with the API
- For idea 1, is defining a fixed evaluation drop (for example a certain amount of percentage lost in the probability of a win) an accurate way to define a "blunder"?
- Idea 2 has two related questions. For Phase II, should we focus on one opening and one trap, or is something like comparing a trap and a gambit better?
- Is a one-player project acceptable for the final project, or should we try to compare a small set of top players instead?
- Which idea do you think best fits the requirements for data?

